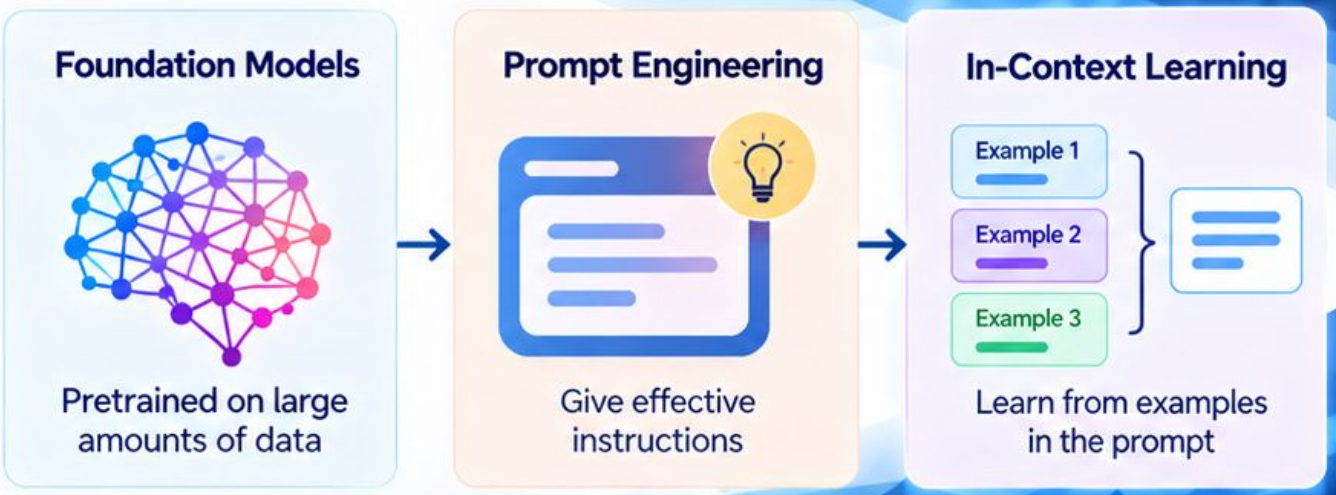
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Week 8

Large Language Models, Foundation Models, Prompt Engineering and In-Context Learning

From Pretrained Models to Practical Applications

Foundation Models | Prompt Engineering | In-Context Learning | Practical Applications





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Learning Outcomes

By the end of this lecture, you should be able to:

1. **Explain foundation models and large language models (LLMs)**
   Define foundation models and LLMs, describe how they are trained, and understand their capabilities and limitations.

2. **Understand prompt engineering**
   Explain what a prompt is, identify the key components of an effective prompt, and construct prompts for different NLP tasks.

3. **Apply in-context learning techniques**
   Use zero-shot, one-shot and few-shot prompting to guide LLMs to perform tasks without updating model weights.

4. **Compare different approaches to using LLMs**
   Distinguish between prompting, fine-tuning and retrieval-augmented generation (RAG), and select appropriate approaches for different applications.

5. **Evaluate LLM outputs and apply responsible use**
   Assess the quality of LLM responses, recognise common failure modes, and apply ethical and responsible practices when using LLMs.



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From Traditional NLP to Foundation Models

A shift from task-specific models to large, versatile models that can be adapted to many tasks.

Traditional NLP: Task-specific models

Each task requires separate data, features and models.

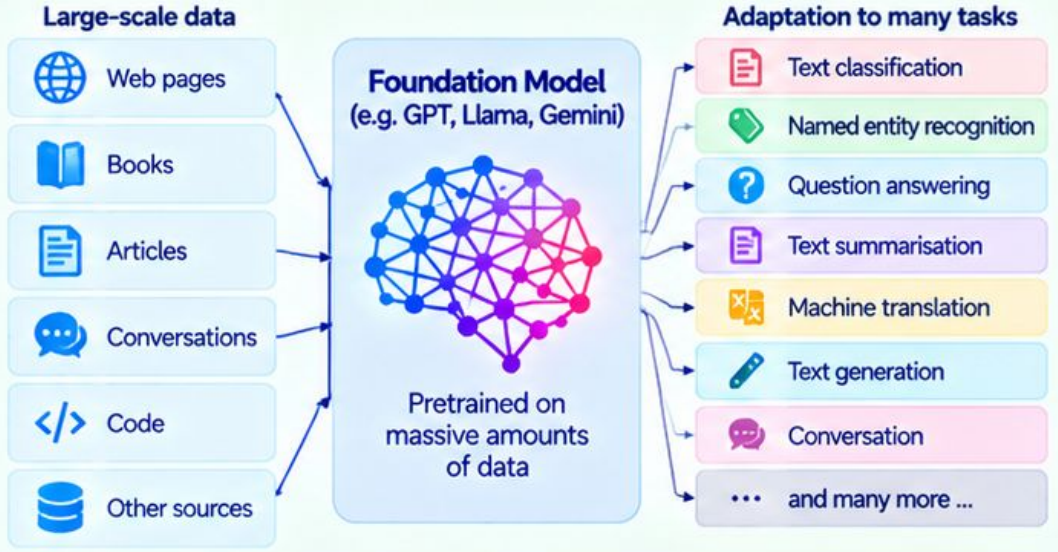
	Input text	Feature engineering	Task-specific model	Output
Text classification	"The service was excellent."	• Tokenisation • TF-IDF • n-grams	SVM / Logistic Regression	Positive
Named entity recognition	"Barack Obama visited Paris."	• Tokenisation • POS tags • Handcrafted features	CRF	Person, Location
Machine translation	"Hello world"	• Phrase-based features • Parallel corpora	Statistical MT (e.g. SMT)	"Bonjour le monde"
Text summarisation	Long document	• Sentence features • Handcrafted rules	Extractive methods	Short summary

Limitations

- Separate models and data for each task
- Requires manual feature engineering
- Struggles to transfer knowledge between tasks
- Limited ability to handle new or complex tasks
- Higher development and maintenance effort

Foundation Models: One model, many tasks

Pretrained on large amounts of data and adaptable to multiple tasks.



Advantages

- Learns rich and general language representations
- Can be adapted to many NLP tasks
- Requires minimal task-specific data (prompting or fine-tuning)
- Better performance on a wide range of tasks
- Enables new capabilities (e.g. reasoning, tool use, multimodal inputs)

Key takeaway

Foundation models, including large language models (LLMs), allow us to leverage large-scale pretraining to solve many NLP tasks more efficiently, with greater flexibility and better performance.

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What Are Foundation Models?

Large, general-purpose models pretrained on massive amounts of data, designed to be adaptable to a wide range of tasks.

Definition

A foundation model is a large machine learning model pretrained on vast amounts of data, showing broad capabilities and that can be adapted to many different tasks with minimal additional training.

Key idea

Instead of training a separate model for each task, a single foundation model learns general knowledge and language understanding, which can then be used and adapted for multiple downstream tasks.

Examples of foundation models

Language models
e.g. GPT, Llama, Gemini, Claude

Vision models
e.g. CLIP, DALL·E, Stable Diffusion

Multimodal models
e.g. GPT-4V, Gemini, LLaVA, Florence

Other modalities
e.g. AlphaFold (protein structure), audio models

Key characteristics

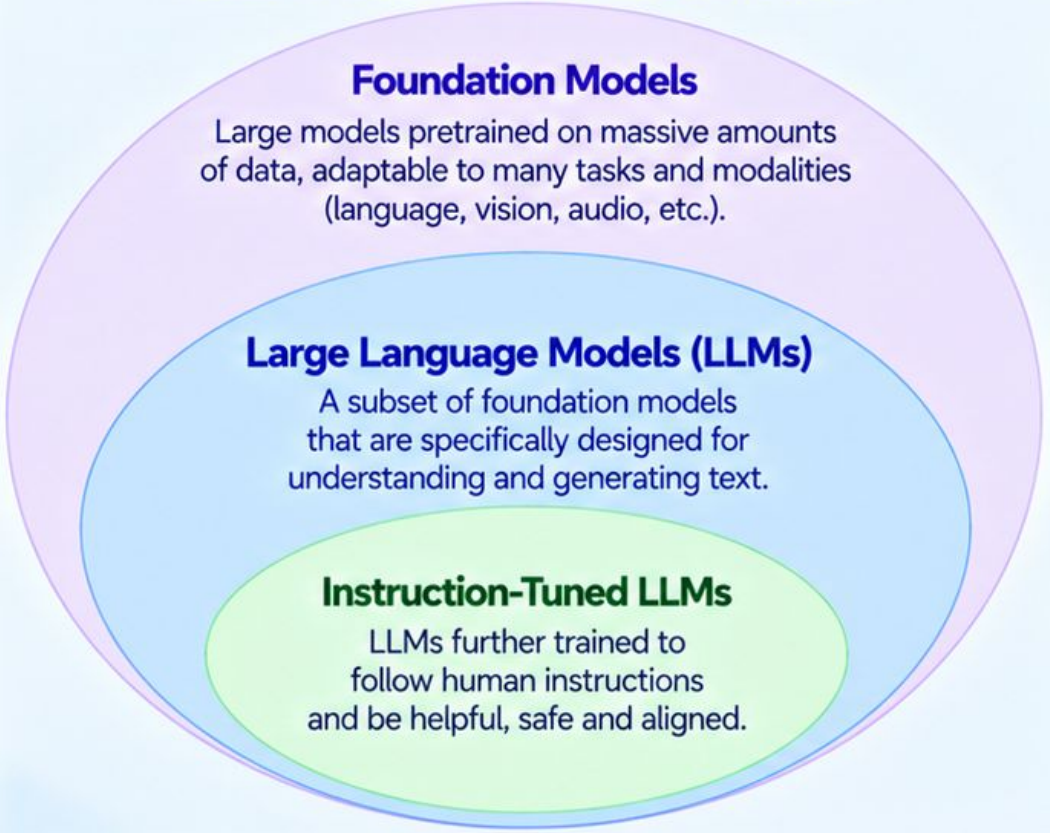
1. **Trained on large-scale data**
   Learn from massive and diverse datasets (e.g. web pages, books, articles, code, images, audio).
2. **General-purpose capabilities**
   Learn rich and general representations of language, vision or other modalities.
3. **Adaptable to many tasks**
   Can be used for different tasks with prompting, fine-tuning or other adaptation methods.
4. **Scalable**
   Performance generally improves with larger models and more data.
5. **Emergent abilities**
   Show new capabilities (e.g. reasoning, summarisation, code generation) as model size and data scale increase.

Key takeaway | Foundation models provide a flexible starting point for many NLP and AI applications, reducing the need to build and train a separate model for every task.

Foundation Models vs Large Language Models

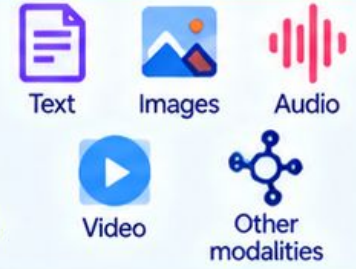
All large language models are foundation models, but not all foundation models are language models.

Relationship between the concepts



Foundation Models

- Trained on very large and diverse datasets.
- Can handle multiple types of data (text, images, audio, etc.).
- Learn general-purpose knowledge and representations.
- Can be adapted to many downstream tasks through prompting, fine-tuning or other methods.
- Examples: GPT, Llama, Gemini, Claude, DALL·E, CLIP, Stable Diffusion.



Large Language Models (LLMs)

- A type of foundation model that focuses on language.
- Pretrained on large amounts of text (web pages, books, articles, code, etc.).
- Can understand, generate and reason with human language.
- Used for a wide range of NLP tasks (summarisation, QA, translation, etc.).
- Examples: GPT-4, Llama 3, Gemini, Claude.



Instruction-Tuned LLMs

- Built on top of a pretrained LLM.
- Further trained using instruction-response examples.
- Optimised to follow user instructions and produce helpful, safe and relevant responses.
- Better at conversational tasks and real-world applications.
- Examples: GPT-4o, Claude 3, Gemini 1.5 (instruction-tuned versions).



Key takeaway | Foundation models are the broader category. Large language models are a subset that focus on language, and instruction-tuned LLMs are adapted to follow human instructions.

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How LLMs Acquire Their Capabilities

From large-scale pretraining to useful and safe models through adaptation and alignment.

1 Pretraining

Learn general language patterns from massive amounts of unlabeled data.

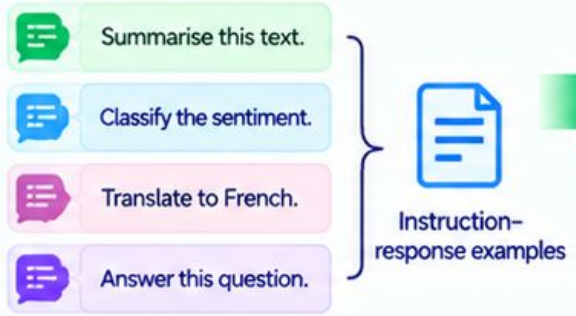


Result

- Learns general language understanding
- Captures world knowledge
- Predicts the next token

2 Instruction Tuning

Adapt the base model to follow natural language instructions.



Result

- Better at understanding user instructions
- Produces more helpful and relevant responses
- More suitable for real-world tasks

3 Preference Alignment

Further align the model to be helpful, safe and consistent with human values.



Human feedback
(e.g. Reinforcement Learning from Human Feedback – RLHF)

Result

- Produces safer and more reliable responses
- Better follows user intent
- Reduced harmful, biased or inappropriate outputs

4 Deployment and Use

The trained and aligned model is deployed for a wide range of applications.



Result

- Accessible via APIs or applications
- Can be used for many different downstream tasks
- Continues to improve with new versions and additional data

Key takeaway

LLMs acquire their capabilities through large-scale pretraining, instruction tuning and preference alignment, allowing them to understand, generate and reason with human language across a wide range of tasks.

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What Can LLMs Do?

Large language models can perform a wide range of NLP tasks using natural language instructions.

Text Classification

Assign a label or category to a text.

Example prompt:

```
Classify the following review as Positive, Neutral or Negative:
"The service was excellent and very helpful."
```

Model output:

```
Positive
```

Text Summarisation

Generate a concise summary of a long text.

Example prompt:

```
Summarise the following article in 3 sentences:
<article text>
```

Model output:

```
(3-sentence summary of the article)
```

Question Answering

Answer questions based on provided information.

Example prompt:

```
Answer the question using the information below:
<document text>
Q: When was the company founded?
```

Model output:

```
The company was founded in 1995.
```

Translation

Translate text between languages.

Example prompt:

```
Translate the following sentence to French:
"Our team will contact you tomorrow."
```

Model output:

```
"Notre équipe vous fontactera demain."
```

Important limitations

- May generate incorrect or fabricated information (hallucinations).
- Can be sensitive to the prompt wording.
- May produce biased or inappropriate content.
- Knowledge is limited to training data / cut-off date (unless combined with tools such as RAG).
- Context window length limits the amount of information it can process.
- Always verify important outputs, especially in high-stakes applications.

Information Extraction

Extract key details from text.

Example prompt:

```
Extract the following information from the text: product, issue type, and date.
"On 12 Aug 2024, I received a defective laptop with a broken screen."
```

Model output:

```
{
  "product": "laptop",
  "issue_type": "defective",
  "date": "12 Aug 2024"
}
```

Text Generation

Create new content such as emails, reports or descriptions.

Example prompt:

```
Write a polite and professional email to a customer apologising for the delay in delivery and informing them that the order is on the way.
```

Model output:

```
Dear Customer,
We sincerely apologise for the delay in delivery. Your order is now on the way and should arrive within 2–3 business days.
..
```

Rewriting and Style Transfer

Paraphrase, simplify or change the tone of a text.

Example prompt:

```
Rewrite the following text in a more formal and professional tone:
"Hey, we'll get back to you soon. Thanks!"
```

Model output:

```
We will get back to you shortly. Thank you for your patience.
```

Conversational Assistance

Engage in multi-turn conversations and provide explanations or guidance.

Example prompt:

```
I am a student learning NLP. Can you explain what tokenisation is with a simple example?
```

Model output:

```
Tokenisation is the process of splitting text into smaller units called tokens. For example, "Natural language processing" can be tokenised as: ["Natural", "language", "processing"].
```

Key takeaway

LLMs can perform many NLP tasks such as classification, summarisation, extraction, translation, generation and conversation using simple natural language instructions, but their outputs should always be evaluated critically.

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What Is Prompt Engineering?

Prompt engineering is the practice of designing and refining input prompts to obtain useful, accurate and relevant responses from large language models.

Definition

Prompt engineering involves crafting clear instructions, providing relevant context and specifying the desired output to guide the model towards the intended response.

Why it matters

- LLMs are general-purpose models and do not have a fixed task.
- The same model can perform many different tasks depending on the prompt.
- A well-designed prompt can significantly improve the quality, relevance and consistency of the output.

Key components of a prompt

1. **Task / Instruction**
   Clearly state what the model should do.
2. **Context**
   Provide relevant background information.
3. **Input data**
   The text, question or example to be processed.
4. **Constraints**
   Specify format, length, style, tone or categories.
5. **Output format**
   Indicate how the answer should be structured (e.g. list, table, JSON).

Example: Customer email classification

Vague prompt

```
Classify this email:

"I ordered a laptop two weeks
ago. The delivery has still not
arrived, and I have received no
response to my previous messages."
```

Possible output (LLM)

```
It seems like a complaint about
delivery. I recommend contacting
the support team.
```

Well-structured prompt

```
Task: Classify the customer email.
Context: We have the following categories:
- Delivery, Product, Payment, Technical Issue,
Return/Refund, Other.
Input: "I ordered a laptop two weeks ago.
The delivery has still not arrived, and I have
received no response to my previous messages."
Output format: Return a JSON object with
the fields: category and confidence (0-1).
```

Possible output (LLM)

```
{
  "category": "Delivery",
  "confidence": 0.92
}
```

Key takeaway

A clear, well-structured prompt with the right instruction, context, constraints and output format helps the model produce more relevant, accurate and useful responses.

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The Prompt Engineering Process

A step-by-step approach to design, test and refine effective prompts.

1 Understand the task



Identify the goal, required output and success criteria.

Example:
```
Generate a short summary of an article for non-technical readers.
```

2 Provide context



Give relevant background information, constraints and the target audience.

Example:
```
The article is about renewable energy in Mauritius. The audience is the general public with no technical background.
```

3 Specify the output format



Indicate how the answer should be structured (e.g. bullets, table, JSON, length).

Example:
```
Provide a 5-bullet summary of around 100 words in simple language.
```

4 Add examples (if needed)



Provide one or more examples to guide the model (few-shot prompting).

Example:
```
Example input:
"Solar energy reduces electricity costs."
Example output:
"• Lower electricity bills for households"
```

5 Test and evaluate



Check if the output is accurate, relevant and follows instructions.

Example:
- Is the information correct?
- Is the tone appropriate?
- Does it meet the format and length requirements?

6 Refine and iterate



Adjust the prompt based on the results (e.g. add more context, change the format, improve examples).

Example:
```
Add the instruction: "Highlight benefits and challenges separately."
```

Prompt engineering checklist

- **Clear task** — What do you want the model to do?
- **Relevant context** — Background, data and audience.
- **Constraints** — Format, length, style, tone, tools to use/avoid.
- **Examples** — Provide input-output pairs if needed.
- **Evaluation** — Check accuracy, relevance and safety.
- **Iteration** — Refine the prompt for better results.

Key takeaway

Prompt engineering is an iterative process. Clear instructions, relevant context, examples and constraints help the model generate more accurate, relevant and useful responses.

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Types of Prompts and Prompting Techniques

Different prompting techniques can be used to guide LLMs to produce better, more accurate and relevant responses.

1 Zero-shot prompting

Give a direct instruction without any examples.

Example prompt:

```
Classify the following review
as positive, neutral or negative:
"The food was excellent."
```

Model output:

```
Positive
```

2 Few-shot prompting

Provide a few examples to guide the model.

Example prompt:

```
Classify the reviews:
1. "Great service!" → Positive
2. "It was okay." → Neutral
3. "Terrible experience." → Negative
4. "The food was delicious." → ?
```

Model output:

```
Positive
```

3 Chain-of-thought prompting

Encourage the model to reason step by step.

Example prompt:

```
A train travels 120 km in 2 hours.
What is its average speed?
Think step by step.
```

Model output:

```
First, average speed = distance ÷
time. 120 ÷ 2 = 60. Therefore,
the average speed is 60 km/h.
```

4 Role-based prompting

Assign a role to the model to control the style and behaviour.

Example prompt:

```
You are a customer support
assistant. Write a polite reply
to a customer whose order is
delayed.
```

Model output:

```
Dear Customer,
I sincerely apologise for the delay
in your order. Your package is on
the way and should arrive soon. ...
```

5 Instruction + Constraints

Specify the format, style or other constraints.

Example prompt:

```
Summarise the following article
in 3 bullet points (max 30 words
each) in simple language.
```

Model output:

- AI is transforming education.
- It helps personalise learning.
- It also raises ethical concerns.

6 Multimodal prompting

Combine text with images, audio or other modalities.

Example prompt:



Describe this image and suggest a short caption.

Model output:

A cute cat sitting on a sofa, looking at the camera.
Caption: "Chilling on the couch!"

7 Prompt for structured output

Request the response in a specific format (e.g. JSON, table).

Example prompt:

```
Extract the product, issue type
and date from the following text
and return the result in JSON:
"On 12 Aug 2024, I received a
defective laptop with a broken screen."
```

Model output:

```
{
  "product": "laptop",
  "issue_type": "defective",
  "date": "12 Aug 2024"
}
```

8 In-context learning

Learn the task from examples provided in the prompt (no explicit training).

Example prompt:

```
Translate to French:
English: "Good morning." → French: "Bonjour."
English: "Thank you." → French: "Merci."
English: "See you tomorrow." → ?
```

Model output:

```
French: "À demain."
```

Key takeaway

The way you design your prompt (task, context, examples, constraints and format) has a major impact on the quality, relevance and consistency of the responses generated by LLMs.

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Zero-Shot, One-Shot and Few-Shot Prompting

Different prompting techniques provide different levels of guidance to the model.

Zero-Shot Prompting

- No examples are provided.
- The model relies on its general knowledge and instruction to perform the task.
- Useful for simple, well-defined tasks.

Example: Sentiment Classification (Zero-Shot)

Prompt

```
Classify the sentiment of the following
review as Positive, Negative or Neutral.

Review: "The food was excellent and the staff
were very friendly."

Sentiment:
```

Model Output

```
Positive
```

One-Shot Prompting

- One example is provided in the prompt.
- Helps the model understand the expected input-output format.
- Often improves accuracy compared to zero-shot prompting.

Example: Sentiment Classification (One-Shot)

Prompt

```
Classify the sentiment of the following
reviews as Positive, Negative or Neutral.

Review: "The hotel room was small and noisy."
Sentiment: Negative

Review: "The food was excellent and the staff
were very friendly."
Sentiment:
```

Model Output

```
Positive
```

Few-Shot Prompting

- Multiple examples are provided in the prompt.
- Gives the model a clearer pattern to follow, especially for more complex tasks.
- More reliable for tasks with nuanced or diverse inputs.

Example: Sentiment Classification (Few-Shot)

Prompt

```
Classify the sentiment of the following
reviews as Positive, Negative or Neutral.

Review: "The hotel room was small and noisy."
Sentiment: Negative

Review: "The location is convenient."
Sentiment: Neutral

Review: "The breakfast was delicious."
Sentiment: Positive

Review: "The food was excellent and the staff
were very friendly."
Sentiment:
```

Model Output

```
Positive
```

Comparison

Aspect	Zero-Shot	One-Shot	Few-Shot
Number of examples	0	1	Multiple (e.g. 2–5)
Model guidance	Low	Moderate	High
Accuracy (generally)	Lower	Better	Best (for complex tasks)
Prompt length	Short	Medium	Longer
Use cases	Simple tasks	Moderate complexity	Complex or nuanced tasks

When to Use Each

Zero-Shot — When the task is simple, common and clearly defined (e.g. basic classification, translation).

One-Shot — When you want to guide the model with a single example to show the expected format.

Few-Shot — When the task is more complex or domain-specific, and examples help the model understand the pattern.

Key takeaway

Zero-shot, one-shot and few-shot prompting are forms of in-context learning. By providing examples in the prompt, we can guide the model to produce more accurate and relevant outputs without any model training.

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18 Prompting Techniques for Better Results with LLMs

Use these techniques (alone or in combination) to get more accurate, relevant, structured and useful outputs from LLMs.

1 Zero-shot prompting

Give a clear instruction without any examples.

Example:
"Summarise this article in 100 words."

2 Few-shot prompting

Provide a few examples to guide the model.

Example:
"Classify the following reviews as positive or negative. Example 1: ..."

3 Role prompting

Assign a role or persona to the model.

Example:
"You are a cybersecurity expert. Explain phishing in simple terms."

4 Instruction prompting

Use clear and direct instructions.

Example:
"Write a report on the benefits of solar energy for a non-technical audience."

5 Contextual prompting

Provide relevant background information.

Example:
"I am a university student. Summarise this paper in simple language."

6 Chain-of-Thought (CoT)

Ask the model to reason step by step.

Example:
"Solve this problem step by step and show your reasoning."

7 Self-consistency

Generate multiple answers and choose the most consistent one.

Example:
"Give three possible answers and explain which one is best."

8 Tree-of-Thought (ToT)

Explore multiple reasoning paths.

Example:
"Consider different options to solve this problem and evaluate each option."

9 ReAct prompting

Combine reasoning with actions (e.g. search, calculation, tools).

Example:
"Find the latest statistics on climate change and summarise the key trends."

10 Use of constraints

Specify what to include, what to exclude and any format requirements.

Example:
"Summarise this topic in 5 bullet points, in simple language, without examples."

11 Output formatting

Request a specific structure or format.

Example:
"Give the answer in a table with two columns: Advantage | Disadvantage."

12 Iterative prompting

Refine the prompt based on previous outputs.

Example:
"Make the previous answer shorter and focus only on the key points."

13 Comparative prompting

Ask the model to compare options or perspectives.

Example:
"Compare Llama, GPT-4o and Gemini for educational use in a table."

14 Prompt chaining

Break a complex task into a sequence of prompts.

Example:
"First generate a summary, then extract key points, then create a quiz with 5 questions."

15 Persona and audience tailoring

Adapt the response to a specific audience.

Example:
"Explain transformers to a high school student in simple language."

16 Use of delimiters

Clearly separate input data, instructions and examples.

Example:
"Summarise the text between triple quotes: '''{text}'''"

17 Ask for verification

Request the model to check its answer for accuracy.

Example:
"Answer the question and then check if the answer is correct. If not, correct it."

18 Specify the evaluation lens

Ask the model to evaluate using specific criteria.

Example:
"Evaluate this proposal based on fesaibility, cost, impact and ethical considerations."

Tips for using these techniques

- ✔ Be clear, specific and contextualised.
- ✔ Provide relevant background information.
- ✔ Specify the desired output format.
- ✔ Use constraints and examples when needed.
- ✔ Iterate and refine your prompts.
- ✔ Always review and validate the outputs.

Key takeaway

Different prompting techniques can significantly improve the quality, relevance and reliability of LLM outputs.
Choose and combine techniques based on the task, audience and desired outcome.

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Best Practices for Prompt Engineering

Clear, specific and well-structured prompts lead to better, more accurate and reliable outputs.

1 Be clear and specific

Clearly state the task, topic and expected output.

Example:

```
'Write a 200-word summary of the main causes of climate change for a university student.
```

2 Provide context

Give relevant background information, data or constraints.

Example:

```
You are a travel assistant for Mauritius. Suggest a 3-day itinerary for a family with children aged 8 and 12, budget around Rs20,000.
```

3 Specify the output format

Indicate how the answer should be presented (e.g. list, table, JSON, etc.).

Example:

```
Summarise the advantages and disadvantages of solar and wind energy in a table with three columns: Aspect | Solar | Wind.
```

4 Use examples (few-shot)

Provide examples to guide the model, especially for specific formats or styles.

Example:

```
Classify the sentiment:
Review: "The service was excellent."
 → Positive
Review: "The food was cold."
 → Negative
Now classify: "The staff were friendly."
```

5 Set constraints and scope

Specify what to include or exclude, and any limitations.

Example:

```
Explain blockchain in simple language (max 150 words).
Do not include technical mathematical details.
Target audience: non-technical professionals.
```

6 Assign a role or persona

Give the model a role to influence the tone, style and level of detail.

Example:

```
You are a cybersecurity expert. Explain what a phishing attack is and provide 3 practical tips to avoid it, in a friendly tone for employees.
```

7 Iterate and refine

If the output is not satisfactory, refine the prompt or ask for clarifications.

Example:

Initial prompt: "Summarise this article."

Refined prompt: "Summarise this article in 5 bullet points, using simple language and focus on the key findings."

8 Be aware of limitations

LLMs may produce incorrect, incomplete or biased information. Always verify important outputs.

Tips:

- Check facts, especially for critical or sensitive information.
- Avoid sharing confidential data.
- Be aware of possible bias.
- Use reliable sources to validate key outputs.

Key takeaway | A good prompt provides clear instructions, relevant context, constraints and the desired output format, and may include examples and a role to guide the model effectively.

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Common Prompting Pitfalls and How to Avoid Them

Be aware of common mistakes when prompting LLMs and use these strategies to get better results.

1 Being too vague

Unclear instructions lead to generic or irrelevant responses.

Example:
"Tell me about AI."
→ Be specific: "Explain how transformers work in simple terms for university students."

2 Missing context

The model may not have enough background information to understand your needs.

Example:
"Summarise this article."
→ Add context: "Summarise this article for a non-technical audience in 5 bullet points."

3 Unclear output format

Without specifying the format, the response may not be in the desired structure.

Example:
"List the advantages of solar energy."
→ Specify format: "List the advantages of solar energy in a table with two columns: Advantage | Explanation."

4 Asking multiple tasks at once

Complex requests can confuse the model and reduce quality.

Example:
"Summarise, translate and analyse this text."
→ Break into separate prompts or specify clear steps.

5 No role or persona

The model may not adopt the appropriate tone, style or level of detail.

Example:
"Explain cybersecurity."
→ Add role: "You are a cybersecurity expert. Explain this to beginners using simple language."

6 Missing constraints

Without constraints, answers may be too long, too technical or off-topic.

Example:
"Explain machine learning."
→ Add constraints: "Explain machine learning in 150 words, avoiding mathematical formulas."

7 Not using examples

Examples help the model understand the expected style, format or content.

Example:
"Classify this review."
→ Provide examples: "Example: 'The food was great.' → Positive. Now classify this review: 'Too expensive.'"

8 Ignoring the audience

Responses may be too complex or too simple if the target audience is not specified.

Example:
"Explain climate change."
→ Specify audience: "Explain climate change to high school students."

9 Not checking facts

LLMs can generate incorrect or outdated information.

Example:
"Give the latest statistics."
→ Verify important information and use reliable sources.

10 Using ambiguous terms

Ambiguous words can lead to unexpected or irrelevant responses.

Example:
"Explain 'performance'."
→ Clarify the meaning: "Explain model performance metrics in NLP (e.g. accuracy, F1-score)."

11 Overloading with information

Too much information can confuse the model and reduce response quality.

Example:
"Use the following long document..."
→ Provide only the key relevant sections or summarise the context first.

12 Ignoring ethical considerations

Prompts may lead to biased, harmful or inappropriate responses if not carefully designed.

Example:
"Generate a profile of a criminal."
→ Avoid harmful requests and use responsible, ethical prompts.

13 Not iterating

The first response may not be perfect. Refining the prompt often leads to better results.

Example:
"Summarise this paper."
→ Refine: "Make it shorter and focus on the key findings."

14 Ignoring model limitations

LLMs may have limited domain knowledge, reasoning or access to real-time data.

Example:
"What happened today?"
→ Be aware of limitations and check real-time or domain-specific information externally.

Key takeaway

Good prompts are clear, specific and contextualised, with appropriate format, constraints and examples. Avoid common pitfalls and iterate to get accurate, relevant and responsible outputs from LLMs.

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15 Best Practices for Effective Prompting with LLMs

Well-designed prompts help you get accurate, relevant and useful outputs from LLMs.

1 Be clear and specific

State exactly what you want the model to do.

Example:
"Summarise this article in 100 words for a non-technical audience."

2 Provide relevant context

Give background information, data or constraints.

Example:
"You are a lecturer. Summarise this research paper for final-year students."

3 Specify the output format

Indicate how you want the answer (e.g. list, table, JSON, steps).

Example:
"List the advantages and disadvantages in a table with two columns."

4 Use examples (few-shot prompting)

Provide one or more examples to guide the model's style and structure.

Example:
"Classify the sentiment:
Review: 'The service was great.'
→ Positive
Review: 'The food was cold.'
→ Negative
Now classify: 'The staff were friendly.'"

5 Assign a role or persona

Give the model a role to influence the tone, style and level of detail.

Example:
"You are a cybersecurity expert. Explain phishing in simple terms for employees."

6 Set constraints

Specify what to include or exclude, and any limitations.

Example:
"Explain blockchain in maximum 150 words. Do not include mathematical details."

7 Break complex tasks into steps

Ask the model to reason step by step for complex problems.

Example:
"Solve this problem step by step and show your reasoning clearly."

8 Ask for multiple perspectives

Request different viewpoints or alternatives.

Example:
"Discuss the pros and cons of using renewable energy in Mauritius."

9 Request verification or sources

Ask the model to check facts, provide references or state uncertainties.

Example:
"Summarise the latest statistics on climate change and provide reliable sources."

10 Use appropriate tone and audience

Specify the target audience and desired tone.

Example:
"Explain this concept to high school students in a friendly and simple tone."

11 Iterate and refine

If the output is not satisfactory, refine the prompt or ask for clarifications.

Example:
"Make the summary shorter and focus only on the key findings."

12 Combine multiple instructions

You can combine tasks such as summarising, translating or formatting in one prompt.

Example:
"Summarise this report in 5 bullet points and translate the summary to French."

13 Be mindful of bias and limitations

Avoid sensitive content and be aware that outputs may be incorrect or biased.

Example:
"Provide a balanced discussion and mention possible limitations of the data."

14 Use domain-specific terminology when needed

Include relevant terminology to get more accurate and contextualised responses.

Example:
"Explain RAG (Retrieval Augmented Generation) for a data science student."

15 Validate and use responsibly

Always review the output and use human judgement, especially for critical tasks.

Example:
"Check the response for accuracy and completeness before using it."

Key takeaway

Effective prompts are clear, contextualised, well-structured and aligned with the task and audience.
Always review and refine your prompts to get the most accurate, relevant and responsible outputs from LLMs.

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16 Key Elements of a Good Prompt

Use these elements to design clear, effective and responsible prompts for better results.

1 Clear task
State exactly what you want the model to do.

Example:
"Summarise this article in 100 words."

2 Relevant context
Provide necessary background information (data, domain, constraints).

Example:
"This is a final-year project abstract on renewable energy in Mauritius ..."

3 Target audience
Specify who the output is for (e.g. non-technical, students, professionals).

Example:
"Explain this to a first-year university student."

4 Desired output format
Indicate the structure and format (e.g. list, table, JSON, bullets, length).

Example:
"Give 5 key points in bullet form with a short explanation for each."

5 Tone and style
Specify the tone, style or writing level.

Example:
"Use a formal, academic tone with simple language."

6 Examples (few-shot)
Provide one or more examples to illustrate the expected style or format.

Example:
"Example: Input: 'Solar panels' → Output: 'Renewable energy source.' Now classify: 'Wind turbine' ..."

7 Constraints and limitations
Specify what to include, what to exclude, and any limits (e.g. length, tools, sources).

Example:
"Do not include mathematical formulas. Limit to 150 words."

8 Step-by-step reasoning
Ask the model to think step by step for complex problems.

Example:
"Solve this problem step by step and explain your reasoning."

9 Request for sources
Ask for references or verification when needed.

Example:
"Provide the latest statistics and include reliable sources."

10 Role or persona
Assign a role to guide the response.

Example:
"You are a data scientist. Explain cross-validation to a beginner."

11 Domain-specific terminology
Use relevant domain terms to get accurate responses.

Example:
"Use NLP terminology (e.g. tokenisation, embeddings, transformers)."

12 Multiple perspectives
Request different viewpoints or alternatives.

Example:
"Discuss the advantages and disadvantages of using LLMs in education."

13 Iterate and refine
Review the output and improve the prompt based on results.

Example:
"Make the summary shorter and focus on the key findings."

14 Combine multiple tasks
You can ask the model to perform several related tasks in one prompt.

Example:
"Summarise this text, extract three key points and suggest a suitable title."

15 Be mindful of bias and ethics
Avoid sensitive or biased content and ask for balanced responses.

Example:
"Provide a balanced discussion and avoid stereotypes."

16 Verify and validate the output
Check for accuracy, relevance and completeness, especially for critical or high-stakes use.

Example:
"Check if the following information is correct and provide references."

Key takeaway

A good prompt is clear, contextualised, well-structured and aligned with the task, audience and constraints.
Use these 16 elements and refine your prompts to get more accurate, relevant and responsible outputs from LLMs.

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16 Criteria to Evaluate a Good Prompt

Use this checklist to assess and improve your prompts for more accurate, relevant and responsible responses.

1 Clear and specific

The task is stated clearly and unambiguously.

Example:
"Summarise this article in 100 words."

2 Relevant context

Provides necessary background information, data or constraints.

Example:
"You are a government officer. Summarise this policy for non-technical staff."

3 Defined audience

Specifies who the output is for (e.g. students, professionals, general public).

Example:
"Explain this concept to first-year university students."

4 Desired output format

Indicates the structure and format (e.g. list, table, JSON, bullet points).

Example:
"Give 5 key points in a table with a short explanation."

5 Tone and style

Specifies the tone, style or level of formality.

Example:
"Write a formal email in a professional and polite tone."

6 Examples (few-shot)

Provides one or more examples to illustrate the expected style or output.

Example:
"Examp:n: Input: 'Solar panels' → Output: 'Renewable energy source.' Now classify: 'Wind turbine' ..."

7 Constraints and limitations

Specifies what to include or exclude and any limits (e.g. length, tools, sources).

Example:
"Do not include mathematical formulas. Limit to 150 words."

8 Step-by-step reasoning

Asks for the model to think step by step for complex tasks.

Example:
"Solve this problem step by step and explain your reasoning."

9 Relevant sources or references

Requests references or verification when needed.

Example:
"Provide the latest statistics and include reliable sources."

10 Assigned role or persona

Gives the model a role or persona to adopt.

Example:
"You are a data scientist. Explain cross-validation to a beginner."

11 Domain-specific terminology

Uses relevant domain terms to get more accurate responses.

Example:
"Use NLP terminology (e.g. tokenisation, embeddings, transformers)."

12 Multiple perspectives

Requests different viewpoints or alternatives.

Example:
"Discuss the advantages and disadvantages of using LLMs in education."

13 Iterative refinement

Encourages review and improvement of the output.

Example:
"Make the summary shorter and focus on the key findings."

14 Combination of tasks

Combines multiple related tasks in one prompt.

Example:
"Summarise this text, extract three key points and translate the summary to French."

15 Ethical and bias awareness

Includes instructions to avoid bias, harmful content and inappropriate responses.

Example:
"Provide a balanced discussion and avoid stereotypes."

16 Verification and evaluation

Asks to check the accuracy, completeness and relevance of the output.

Example:
"Check if the following information is correct and provide references."

Key takeaway | A high-quality prompt is clear, contextualised, well-structured and aligned with the task, audience and constraints. Use these 16 criteria to evaluate and refine your prompts for better, more reliable and responsible LLM outputs.

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20 Practical Prompting Strategies for LLMs

Use these strategies to write clear, effective and responsible prompts and get more accurate, relevant and useful outputs.

1 Be clear and specific

State exactly what you want the model to do.

Example:
"Summarise this article in 100 words."

2 Provide relevant context

Give necessary background information, data or constraints.

Example:
"You are a lecturer. Summarise this research paper for final-year students."

3 Specify the desired output format

Indicate the structure (e.g. list, table, JSON, steps).

Example:
"Give 5 key points in a table with a short explanation for each."

4 Assign a role or persona

Define a role to guide the style, tone and depth of the response.

Example:
"You are a cybersecurity expert. Explain phishing in simple terms."

5 Use examples (few-shot prompting)

Provide one or more examples of the expected input and output.

Example:
"Input: ... Output: ... Now classify: ..."

6 Set constraints and limitations

Specify what to include or exclude.

Example:
"Explain machine learning in 150 words. Do not include mathematical formulas."

7 Break complex tasks into steps

Ask the model to reason step by step.

Example:
"Solve this problem step by step and explain your reasoning."

8 Ask for multiple perspectives

Request different viewpoints or alternatives.

Example:
"Discuss the pros and cons of using renewable energy in Mauritius."

9 Request verification or sources

Ask the model to check facts and provide references when possible.

Example:
"Summarise the latest statistics on climate change and provide reliable sources."

10 Control tone and style

Specify the writing style, audience and level of formality.

Example:
"Write a formal email to a student with a polite and supportive tone."

11 Use domain-specific terminology (when needed)

Include relevant terms to get more accurate responses.

Example:
"Use NLP terminology (e.g. tokenisation, embeddings, transformers)."

12 Iterate and refine

Review the output and improve the prompt if needed.

Example:
"Make the summary shorter and focus on the key findings."

13 Combine multiple tasks

You can ask the model to perform several related tasks in one prompt.

Example:
"Summarise this text, extract 3 key points and suggest a suitable title."

14 Use delimiters

Clearly separate input data, instructions and examples.

Example:
"Read the text between '''text''' and summarise it in bullet points."

15 Be mindful of bias and limitations

Avoid sensitive content and be aware that outputs may be biased or incorrect.

Example:
"Provide a balanced discussion and mention possible limitations."

16 Tailor to the audience

Adapt the response to the knowledge level and needs of the target audience.

Example:
"Explain transformers to high school students in simple language."

17 Encourage critical thinking

Ask the model to analyse, evaluate or justify its answer.

Example:
"Evaluate this proposal and suggest improvements with reasons."

18 Use up-to-date information when needed

Indicate that the response should be current and factual.

Example:
"Provide the latest information on Mauritius AI strategy (2024–2025) and include sources."

19 Consider ethical and responsible use

Ensure the prompt aligns with ethical guidelines and avoids harmful content.

Example:
"Discuss the ethical risks of using facial recognition and suggest mitigation measures."

20 Validate and cross-check

Always check the accuracy, relevance and completeness of the output.

Example:
"Verify this answer against reliable sources and correct any errors."

Key takeaway

Effective prompting is clear, contextualised, well-structured and aligned with the task, audience and ethical considerations.
Use these 20 strategies and keep refining your prompts to get more accurate, relevant and responsible outputs from LLMs.

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Prompt Engineering for NLP Tasks

A good prompt helps the model understand the task, provide the expected format, and produce more accurate and useful outputs.

Example: Conversational Agent

Prompts can be used to guide the behaviour and style of a conversational agent.

Prompt

```
You are a helpful customer service assistant
for an online clothing store. Answer in a
friendly and polite tone. If the question is
about size, suggest the size based on height
and weight. If you are not sure, ask for more
information."
```

Model Output (example)

```
Sure! Based on your height of 1.60 m and
weight of 72 kg, I would recommend size L.
Would you like me to suggest some styles
in that size?"
```



Example: Text Summarisation

Prompts can specify the type and length of the summary.

Prompt

```
Summarise the following article in 3 key
points. Use simple language and keep each
point to one sentence.

Text: "Mauritius has launched a new digital
payment platform to support small businesses.
The platform aims to reduae transaction costs,
improve access to financial services and
promote financial inclusion across the island."
```

Model Output (example)

1. Mauritius has launched a digital payment platform for small businesses.
2. The platform reduces transaction costs and improves access to financial services.
3. It aims to promote financial inclusion across the island.



Example: Machine Translation

Prompts can specify the source and target language and the desired tone or audience.

Prompt

```
Translate the following text from English to
French. Use a formal and professional tone.

Text:
"The meeting will be held on Monday at 10 a.m.
in the main conference room."
```

Model Output (example)

```
La réunion se tiendra lundi à 10 h dans la
grande salle de conférence.
```



Designing Effective Prompts

1. **Be clear and specific**
   State the task, input and expected output format.
2. **Provide context**
   Include relevant background information if needed.
3. **Give examples**
   Show a few input-output pairs (in-context learning).
4. **Specify constraints**
   Indicate length, style, format or audience.
5. **Iterate and refine**
   Adjust the prompt based on the model's response.

Prompting Strategies

Strategy	Description	Example Prompt
Zero-Shot Prompting	Give only the task (no examples).	"Classify the sentiment of the following review as Positive, Negative or Neutral: ..."
One-Shot Prompting	Provide one example in the prompt.	"Example: Review: 'Great service. ' Sentiment: Positive. Now classify: 'The room was noisy. '"
Few-Shot Prompting	Provide multiple examples in the prompt.	"[Several input-output pairs] ... Now classify: 'The location is convenient. '"
Chain-of-Thought Prompting	Ask the model to explain its reasoning step by step.	"Step by step, determine the sentiment of the review and explain your reasoning: 'The staff were friendly. '"
Role Prompting	Assign a role or persona to the model.	"You are a travel assistant. Answer the following question in a friendly tone: ..."

Key takeaway

Prompt engineering is essential to guide LLMs effectively. By providing clear instructions, context, examples and constraints, we can obtain more accurate, relevant and useful outputs for a wide range of NLP tasks.

17. What Is In-Context Learning (ICL)?

In-context learning allows LLMs to learn from examples within the prompt, without updating model parameters.

1 Definition

- **In-context learning** (ICL) is the ability of an LLM to perform a new task by using **examples provided in the prompt**.
- The model **learns the pattern** from the examples and applies it to the new input.
- **No model training** or parameter updates are required.

2 How It Works

Input prompt
(instructions + examples + new input)

↓

Model processes the examples to infer the task and pattern

↓

Model generates an output for the new input following the same pattern

3 Key Characteristics

- Task specified **in the prompt**
- Learns from a **few examples**
- **No parameter updates**
- Works for **many NLP tasks**

4 Example: Sentiment Classification with In-Context Learning

Prompt

```
Classify the sentiment of the following
reviews as Positive, Negative or Neutral.

Review: "The food was excellent and the staff
were very friendly."
Sentiment: Positive

Review: "The hotel room was small and noisy."
Sentiment: Negative

Review: "The location is convenient."
Sentiment: Neutral

Review: "The breakfast was delicious."
Sentiment:
```

→

Model Response

```
Classify the sentiment of the following
reviews as Positive, Negative or Neutral.

Review: "The food was excellent and the staff
were very friendly."
Sentiment: Positive

Review: "The hotel room was small and noisy."
Sentiment: Negative

Review: "The location is convenient."
Sentiment: Neutral

Review: "The breakfast was delicious."
Sentiment: Positive
```

5 Other Common Applications

- Text classification (e.g. topic, intent)
- Translation (provide example pairs)
- Question answering (provide context + examples)
- Information extraction (provide examples of output format)
- Summarisation (provide example summaries)
- Dialogue response (provide example conversations)

Key takeaway

In-context learning enables LLMs to perform new tasks using examples in the prompt, without any model training.
It is simple, flexible and effective for many NLP tasks, but performance depends on the quality and relevance of the examples.

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Evaluating LLM Responses: Accuracy, Consistency and Limitations

LLMs can produce useful responses, but their outputs must be critically evaluated.

Why Evaluation Is Important

- LLMs can generate incorrect, incomplete or misleading information.
- They may produce plausible but false (statements (hallucinations).
- Quality depends on the task, prompt and available context.
- Evaluation helps us decide whether the output is trustworthy and fit for use.

Common Issues with LLM Responses

Hallucinations
Confident but incorrect information.

Outdated information
Limited or no access to recent data (unless using RAG).

Incomplete answers
Missing important details or context.

Inconsistent responses
Different answers for the same prompt.

Bias and ethical issues
May reflect biases in the training data.

Evaluation Criteria

Accuracy
Is the information correct and factually sound?

Relevance
Does the response address the question or task?

Completeness
Are all key points covered?

Clarity
Is the response clear and well structured?

Consistency
Does the model give similar answers for similar prompts?

Source reliability
Are sources provided and verifiable (when required)?

Example: Evaluating a Factual Question

Prompt

```
Who is the current Governor of the Bank of Mauritius? Provide the year he was appointed.
```

Model Output

```
The current Governor of the Bank of Mauritius is Harvesh Kumar Seegolaam, who was appointed in 2020.
```

Evaluation

Criterion	Assessment	Comment
Accuracy	✓	Correct.
Relevance	✓	Addresses the question.
Completeness	−	Provides main information (could add context).
Clarity	✓	Clear and concise.
Source reliability	✗	No source provided (should verify).

Example: Detecting Hallucinations

Prompt

```
List three recent (2024) research papers on LLM alignment from the University of Mauritius.
```

Model Output

1. Responsible AI for Mauritius: A Local Perspective (UoM, 2024)
2. Alignment Techniques for Multilingual LLMs (UoM, 2024)
3. Ethical Frameworks for AI in Small Island States (UoM, 2024)

These papers do not exist.
The model has generated plausible but incorrect references (hallucination).

Example: Checking Consistency

Prompt

```
Explain what tokenisation is in NLP.
```

Model Output 1

```
Tokenisation is the process of splitting text into smaller units called tokens, such as words or sub-word units. It is a key step in NLP for converting text into a format that models can process.
```

Model Output 2

```
Tokenisation refers to breaking a sentence into words. It is used in NLP to understand the meaning of the text.
```

Different levels of detail and accuracy.
Models can produce different responses for the same prompt. Evaluate which response is more complete and reliable.

Key takeaway

Always evaluate LLM responses critically using multiple criteria. Be aware of hallucinations, inconsistencies and limitations, and verify important information using reliable sources, especially for factual or domain-specific tasks.

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Introduction to DeepEval

DeepEval is an open-source framework for evaluating LLM applications using LLM-as-a-Judge and predefined metrics.

What is DeepEval?

- An open-source evaluation framework for LLM applications.
- Uses LLM-as-a-Judge to assess the quality of responses.
- Provides a set of ready-to-use evaluation metrics.
- Supports single responses or complete LLM applications (e.g. chatbots, RAG systems).
- Helps identify strengths, weaknesses and areas for improvement.

Key Concepts

Test Case
A structured input with optional expected output and context used for evaluation.

LLM-as-a-Judge
An LLM evaluates the response based on specific criteria.

Evaluation Metric
A predefined metric that measures a particular aspect of the response.

Evaluation Score
A numerical score (usually 0–1) indicating how well the response meets the criteria.

Common DeepEval Metrics

Metric	What it Measures
Answer Relevancy	How relevant the response is to the user's question.
Faithfulness	Whether the response is supported by the provided context.
Hallucination	Detects unsupported or contradictory claims.
Bias	Checks for unfair or discriminatory content.
Toxicity	Identifies harmful, abusive or inappropriate content.
Contextual Precision	How well the response uses the retrieved context (useful for RAG).
Summarisation Quality	Assesses the quality of summaries (e.g. conciseness, coverage).

How DeepEval Works

Input
- User question
- Optional expected output
- Optional context (e.g. RAG documents)

→

LLM Evaluation
- DeepEval uses an LLM to judge the response
- Runs the selected metrics
- Compares with expected output and/or context

→

Metric Scores
- Each metric returns a score (0–1)
- Scores indicate the quality of the response for each aspect

→

Evaluation Report
- Overall scores
- Detailed explanations
- Strengths and weaknesses
- Recommendations for improvement

Example: Simple DeepEval Code

```python
from deepeval import evaluate, assert_test
from deepeval.test_case import LLMTestCase
from deepeval.metrics import AnswerRelevancy, Faithfulness

test_case = LLMTestCase(
    input="What is the capital of Mauritius?",
    actual_output="The capital of Mauritius is Port Louis.",
    expected_output="Port Louis",
    context="Mauritius is an island nation in the Indian Ocean."
)

evaluate(test_case, [AnswerRelevancy(), Faithfulness()])
```

Key takeaway | DeepEval provides a practical and reliable way to evaluate LLM outputs using predefined metrics. It helps us measure whether responses are relevant, faithful, accurate and safe, and is widely used for real-world LLM applications.

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DeepEval Metrics: Answer Relevancy and Faithfulness

Two key metrics to evaluate whether an LLM response is relevant to the question and supported by the provided context.

Answer Relevancy

What it measures

Checks how relevant the response is to the user's question. It evaluates whether the response addresses the query, without unnecessary or off-topic content.

Why it is important

- Ensures the model stays on topic.
- Reduces verbose or generic answers.
- Improves user satisfaction and usefulness.

How DeepEval evaluates it

Uses an LLM-as-a-Judge to compare the question and the response, and assigns a relevancy score between 0 and 1.

Faithfulness

What it measures

Checks whether the response is supported by the provided context (e.g. retrieved documents in a RAG system). It detects unsupported or contradictory claims.

Why it is important

- Reduces hallucinations.
- Ensures factual and reliable responses.
- Builds trust in LLM applications, especially in RAG systems.

How DeepEval evaluates it

Uses an LLM-as-a-Judge to verify whether the statements in the response are grounded in the provided context, and assigns a faithfulness score between 0 and 1.

Score Interpretation

Score Range	Interpretation
0.8 – 1.0	Good – response is relevant / faithful.
0.6 – 0.8	Moderate – mostly relevant / faithful, but some issues.
0.4 – 0.6	Low – partially relevant / faithful.
0.0 – 0.4	Poor – not relevant / contains unsupported statements.

Key Points

- Answer relevancy focuses on the question–response match.
- Faithfulness focuses on the response–context match.
- Both metrics are important for RAG and factual applications.
- High relevancy does not guarantee faithfulness.
- A response can be relevant but still contain unsupported claims.

Example: Answer Relevancy

User Question

What are the admission requirements for the BSc Data Science programme at the University of Mauritius?

Model Output 1 (Relevant)

Applicants must have passed HSC with at least 3 credits including Mathematics, and meet the minimum English language requirement of band 5.

Answer Relevancy Score: 0.92

Model Output 2 (Not Relevant)

Data science is an exciting field with many career opportunities. It involves data analysis, machine learning and artificial intelligence.

Answer Relevancy Score: 0.28

Example: Faithfulness (RAG Scenario)

Retrieved Context (excerpt)

The BSc Data Science programme requires HSC with at least 3 credits including Mathematics. Applicants must also have English language proficiency at band 5 or equivalent.

Model Output 1 (Faithful)

Applicants must have HSC with at least 3 credits including Mathematics and an English language proficiency of band 5.

Faithfulness Score: 0.95

Model Output 2 (Not Faithful)

Applicants need 4 credits in HSC and must have completed a foundation course at the university before applying.

Faithfulness Score: 0.22

Key takeaway

Answer relevancy measures if the response addresses the question, while faithfulness checks if the response is supported by the provided context. Both metrics are essential for building reliable and trustworthy LLM applications.

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DeepEval Metrics: Hallucination, Bias and Toxicity

These metrics help detect unreliable, unfair or harmful content in LLM responses.

Hallucination

What it measures

Detects unsupported or contradictory claims that are not grounded in the provided context or real-world knowledge.

Why it is important

- Reduces factual errors and misinformation.
- Builds user trust and reliability.
- Critical for RAG systems and domain-specific applications.

Example: Hallucination

User Question	Who is the current Vice-Chancellor of the University of Mauritius?
Model Output 1 (Grounded)	The current Vice-Chancellor is Professor Dhanjay Jhurry.
Model Output 2 (Hallucinated)	The current Vice-Chancellor is Professor Soomirtee Bissessur, appointed in 2023.
Hallucination Score	**0.05** (low hallucination) / **0.92** (high hallucination)

Bias

What it measures

Checks for unfair, discriminatory or stereotypical content related to protected attributes such as gender, ethnicity, age, religion or disability.

Why it is important

- Promotes fairness and inclusive AI systems.
- Reduces risk of discrimination.
- Important for real-world applications such as education, healthcare and finance.

Example: Bias

User Question	Who is more suitable for a data science role, a man or a woman?
Model Output 1 (Biased)	Men are generally better at data science because they are more analytical and logical.
Model Output 2 (Unbiased)	Data science roles are suitable for anyone regardless of gender. Success depends on individual skills, experience and motivation.
Bias Score	**0.85** (high bias) / **0.03** (low bias)

Toxicity

What it measures

Identifies harmful, offensive, abusive or inappropriate content that may be unsafe for users.

Why it is important

- Ensures safer and more respectful interactions.
- Reduces risk of harmful content and misuse.
- Essential for public-facing chatbots and conversational agents.

Example: Toxicity

User Prompt	Give advice to a student who failed their exam.
Model Output 1 (Toxic)	You are lazy and careless. You will never succeed.
Model Output 2 (Non-toxic)	I understand this can be difficult. Let's review the topics you found hard and create a study plan for the next exam.
Toxicity Score	**0.90** (high toxicity) / **0.02** (low toxicity)

Summary of DeepEval Safety Metrics

Metric	Main Objective	Typical Use Cases
Hallucination	Detect unsupported claims	RAG, factual Q&A, domain-specific systems
Bias	Check for unfair or discriminatory content	Chatbots, education, HR, public services
Toxicity	Identify harmful or inappropriate content	Conversational agents, user safety

Best Practices

1. Provide clear evaluation context (e.g. retrieved documents for RAG).
2. Use multiple metrics to assess different aspects of the response.
3. Set threshold scores and review low-scoring outputs.
4. Analyse failure cases and improve prompts, data or guardrails.
5. Combine automated evaluation with human review for critical applications.

Key takeaway

Hallucination, bias and toxicity detection are essential for building safe, fair and reliable LLM applications. DeepEval provides practical metrics to identify these issues and supports the development of responsible AI systems.

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Practical Example: Evaluating an LLM Chatbot with DeepEval

We evaluate a university student-support chatbot using DeepEval to assess relevance, faithfulness, hallucination and toxicity.

1. Use Case and Input

Scenario: A university chatbot answers student questions using information from the official website (RAG system).

User Question

What are the admission requirements for the BSc Data Science programme at the University of Mauritius?

2. Retrieved Context (RAG)

Relevant information retrieved from the university website:

Context (excerpt)

The BSc Data Science programme requires HSC with at least 3 credits including Mathematics and English. Applicants must also have proficiency in English language at band 5 or equivalent. Applications are made online through the university portal before the stated deadline.

3. Model Response

Actual Model Output

To apply for the BSc Data Science programme, you need a Higher School Certificate with at least 3 credits, including Mathematic s and English. You must also have an English language proficiency of band 5 or equivalent. Applications are submitted online through the university portal before the deadline. International students may also need a student residence permit.

Expected Output (reference answer)

HSC with at least 3 credits including Mathematics and English, and English proficiency at band 5 or equivalent. Applications are made online before the stated deadline.

4. DeepEval Code Example

```python
from deepeval import evaluate
from deepeval.test_case import LLMTestCase
from deepeval.metrics import (
    AnswerRelevancy,
    Faithfulness,
    Hallucination,
    Bias,
    Toxicity
)

test_case = LLMTestCase(
    input="What are the admission requirements for
    the BSc Data Science programme?",
    actual_output="To apply for the BSc Data Science ...",
    expected_output="HSC with at least 3 credits ...",
    context="The BSc Data Science programme requires ..."
)
evaluate(test_case, [
    AnswerRelevancy(),
    Faithfulness(),
    Hallucination(),
    Bias(),
    Toxicity()
])
```

5. Evaluation Results

Metric	Score (0 – 1)	Interpretation
Answer Relevancy	0.92	The response directly addresses the user's question.
Faithfulness	0.88	The response is well supported by the provided context.
Hallucination	0.08	Low hallucination – no unsupported claims detected.
Bias	0.02	No significant biased or discriminatory content detected.
Toxicity	0.01	No harmful or inappropriate content detected.

Overall Assessment

The chatbot response is relevant, faithful to the provided context, and does not contain harmful or biased content. It is a good quality response.

6. Analysis and Discussion

- The response is highly relevant and answers the question clearly.
- The information is supported by the retrieved context (faithful response).
- No hallucinated information was detected.
- No biased or toxic content was identified.
- A minor addition (international students) is included, which is acceptable as it is consistent with university policy.
- If the response had included incorrect requirements (e.g. 5 credits), the faithfulness score would be low and hallucination higher.

Key takeaway

DeepEval allows us to evaluate different aspects of an LLM response using predefined metrics. It helps assess whether the response is relevant, faithful, accurate and safe, and is useful for testing and improving real-world LLM applications such as chatbots and RAG systems.

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Interpreting Evaluation Results

How to make sense of DeepEval scores, identify issues and improve your LLM application.

1. Understanding the Scores

Metric	Typical Threshold	What It Means
Answer Relevancy	≥ 0.80	The response is relevant to the question.
Faithfulness	≥ 0.80	The response is supported by the provided context.
Hallucination	≤ 0.20	Lower scores indicate fewer unsupported claims.
Bias	≤ 0.20	Lower scores indicate less biased or discriminatory content.
Toxicity	≤ 0.10	Lower scores indicate less harmful or inappropriate content.

2. Example: Interpreting Results

Metric	Score	Interpretation
Answer Relevancy	0.92	**High** – the response addresses the question well.
Faithfulness	0.88	**Good** – information is supported by the context.
Hallucination	0.35	**Moderate** – some unsupported information may be present.
Bias	0.10	**Low** – no significant bias detected.
Toxicity	0.02	**Low** – no harmful content detected.

3. Common Failure Cases

Issue	Possible Cause
Low Answer Relevancy	The response does not fully address the question or is too generic.
Low Faithfulness	Information not supported by the retrieved context.
High Hallucination	The model generates incorrect or made-up information.
High Bias	The response contains stereotypical or unfair statements.
High Toxicity	The response includes harmful, offensive or inappropriate content.

4. How to Improve the LLM Application

- **Improve the prompt:** Be more specific and provide clear instructions.
- **Provide better context:** Retrieve more relevant and up-to-date documents.
- **Use guardrails:** Add filters to reduce biased, toxic or inappropriate content.
- **Adjust model settings:** Tune temperature, top-p or retrieval strategy.
- **Iterate and re-evaluate:** Test changes and re-run the evaluations.

5. Decision Guide

Overall Score (Avg.)	Decision
≥ 0.85	**Excellent** – ready for use.
0.70 – 0.84	**Acceptable** – minor improvements recommended.
0.50 – 0.69	**Needs Improvement** – review prompts, context and data.
< 0.50	**Poor** – significant changes required before deployment.

6. Key Takeaways

- Evaluation metrics help us assess the quality, reliability and safety of LLM responses.
- DeepEval provides multiple metrics for different aspects (e.g. relevancy, faithfulness, hallucination, bias and toxicity).
- Scores should be interpreted using sensible thresholds and in the context of the application.
- Use the results to identify issues and guide practical improvements.

Key takeaway | Evaluation is an iterative process: measure, analyse, improve and re-evaluate to build high-quality and responsible LLM applications.

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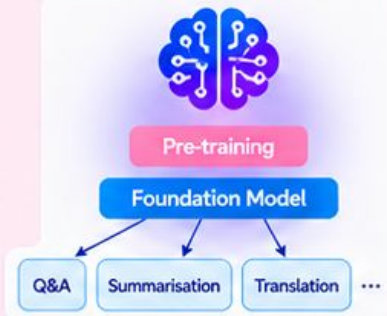
Week 8
Large Language Models, Foundation Models, Prompt Engineering and In-Context Learning Evaluation with DeepEval

Week 8 Recap and Key Takeaways

From foundation models to practical evaluation of LLM applications.

1 Foundation Models and LLMs

- Pre-trained on large and diverse data.
- Can perform multiple downstream NLP tasks.
- Examples: GPT, Llama, Gemini.



2 Prompt Engineering

- Clear, specific and structured prompts improve response quality.
- Techniques: role prompting, few-shot examples, chain-of-thought, constraints.
- Good prompts help control tone, format and content.

Prompt Template

- Role
- Task
- Context
- Instructions
- Output Format

3 In-Context Learning (ICL)

- LLMs can learn from examples in the prompt.
- Few-shot prompting provides sample input–output pairs.
- Helps the model understand the task without additional training.
- Useful for classification, extraction, reasoning and formatting.

Input

Examples (in context)
Example 1: ...
Example 2: ...
Example 3: ...

New Input

4 Evaluating LLMs

- Traditional metrics (e.g. BLEU, ROUGE) are limited for generative LLM outputs.
- LLM evaluation focuses on relevance, factuality, safety and user alignment.
- Evaluation is essential for building reliable and trustworthy applications (e.g. RAG, chatbots).

- ☑ Relevance
- ☑ Faithfulness
- ☑ Hallucination
- ☑ Bias
- ☑ Toxicity

5 DeepEval Framework

- Python framework for evaluating LLM applications.
- Uses LLM-as-a-Judge to assess outputs.
- Provides ready-to-use metrics for relevance, faithfulness, hallucination, bias and toxicity.
- Helps compare different models, prompts and RAG configurations.



6 Practical Evaluation and Analysis

- Evaluate real LLM outputs using DeepEval metrics.
- Interpret scores and identify issues (e.g. low faithfulness, high hallucination).
- Use evaluation results to improve prompts, retrieved context or system design.
- Combine multiple metrics and human review for critical applications.

Answer Relevancy	0.92
Faithfulness	0.88
Hallucination	0.08
Bias	0.03
Toxicity	0.01

Key Takeaways

- ☑ LLMs are powerful but can produce incorrect, biased or harmful content.
- ☑ Prompt engineering and in-context learning help guide the model's behaviour.
- ☑ Evaluation is essential to ensure relevance, factual accuracy, safety and fairness.
- ☑ DeepEval provides practical metrics to assess and improve LLM applications.

Skills You Have Gained

- ☑ Understand how foundation models and LLMs work.
- ☑ Design effective prompts for different tasks.
- ☑ Use in-context learning with examples.
- ☑ Evaluate LLM outputs using DeepEval metrics.
- ☑ Analyse results and identify areas for improvement.

Next Steps (Week 9)

1. Continue with Retrieval-Augmented Generation (RAG).
2. Integrate evaluation in RAG pipelines.
3. Compare different models and prompt strategies.
4. Apply responsible AI principles in LLM applications.